

Risk, Sufficiency, Completeness, and Ancillarity

The initial section of this chapter develops a basic framework for inference. Later sections concern the notion of sufficiency that arises when data can be summarized without any loss of information.

3.1 Models, Estimators, and Risk Functions

Inferential statistics can be viewed as the science or art of learning about an unknown parameter θ from data X . For most applications, X will be a random vector. The parameter θ may be a single constant, but more commonly takes values in some subset of $\mathbb{R}^p$. The parameter θ and data X are related through a model in which the distribution of X is determined by θ . The distribution when the parameter is θ is denoted P_θ , and we write

$$X \sim P_\theta.$$

Formally, a model should be a mapping $\theta \rightsquigarrow P_\theta$, but more commonly, a model is written as the set of distributions for X , $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$, where the parameter space Ω is the set of all possible values for θ .

A statistic is a function of the data X and can be viewed as providing partial information about the data when the function is many-to-one. A typical example, when X is a random vector in $\mathbb{R}^n$, would be the sample average

$$\delta(X) = \overline{X} = \frac{X_1 + \cdots + X_n}{n}.$$

In a broad sense there are two major categories for inference: estimation and hypothesis testing. For now we focus on estimation, in which the goal is to find statistic δ so that $\delta(X)$ is close to $g(\theta)$. Then δ or $\delta(X)$ is called an estimator of $g(\theta)$. The case $g(\theta) = \theta$ is allowed, and, when θ is a vector, $g(\theta) = \theta_k$ may be a fairly typical situation of interest. Hypothesis testing is introduced and studied in Chapter 12.

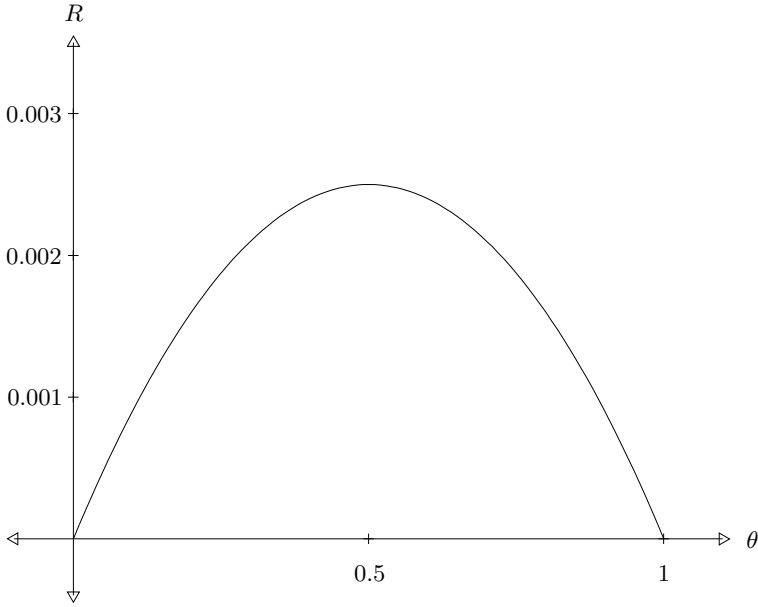


Fig. 3.1. Binomial risk function: $R(\theta, \delta)$.

Example 3.1. For a coin toss, the chance of heads is very close to $1/2$. Suppose instead we stand a coin on its edge, balancing it with a finger on top, and spin it by flicking it with a different finger. If this is done 100 times, with the trials independent and a common chance θ of heads on each spin, then the total number of heads X should have a binomial distribution. Thus

$$X \sim \text{Binomial}(100, \theta).$$

Viewing X as our data and taking

$$P_\theta = \text{Binomial}(100, \theta), \quad \theta \in [0, 1] = \Omega,$$

our model $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is the set of binomial distributions with 100 trials.

In this example, a natural estimator of θ is $\delta(X) = X/100$, the proportion of heads in 100 spins. In the sequel we study the performance of estimators, trying to decide when an estimator, such as δ here, is good or optimal in some sense. An adequate answer to these questions must involve criteria that judge the performance of estimators. One standard approach to making such judgments is called *decision theory*. For estimation, this approach begins with a loss function L chosen so that $L(\theta, d)$ is the loss associated with estimating $g(\theta)$ by a value d . It is natural to assume $L(\theta, g(\theta)) = 0$, so that there is no loss for the correct answer, and $L(\theta, d) \geq 0$, for all θ and d . Because X is

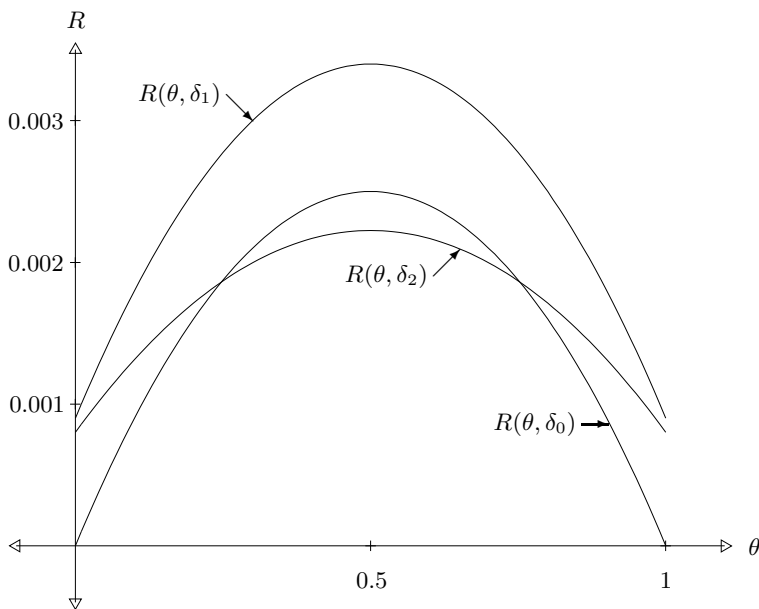


Fig. 3.2. Risks for δ_0 , δ_1 , and δ_2 .

random, $L(\theta, \delta(X))$ is random and can be large if we are unlucky, even if δ is an excellent estimator. Accordingly, an estimator δ is judged by its average loss or *risk function* R , defined as

$$R(\theta, \delta) = E_{\theta} L(\theta, \delta(X)).$$

Here E_{θ} denotes expectation when $X \sim P_{\theta}$.

Example 3.1, continued. Suppose $X \sim \text{Binomial}(100, \theta)$, $\delta(X) = X/100$, $g(\theta) = \theta$, and the loss function is given by $L(\theta, d) = (\theta - d)^2$, called squared error loss. Then the risk function for δ is

$$R(\theta, \delta) = E_{\theta}(\theta - X/100)^2 = \frac{\theta(1 - \theta)}{100}, \quad \theta \in [0, 1].$$

A plot of this risk function is given in [Figure 3.1](#).

A fundamental problem arises when one compares estimators using risk functions: if the risk functions for two estimators cross, there is no clear decision which estimator is best. For instance, in our binomial example, if $\delta_0(X)$ is the original estimator $X/100$, $\delta_1(X) = (X+3)/100$, and $\delta_2(X) = (X+3)/106$, then $R(\theta, \delta_0) = \theta(1 - \theta)/100$, $R(\theta, \delta_1) = (9 + 100\theta(1 - \theta))/100^2$, and $R(\theta, \delta_2) = (9 - 8\theta)(1 + 8\theta)/106^2$. These functions are plotted together in [Figure 3.2](#). Looking at the graph, δ_0 and δ_2 are both better than δ_1 , but the comparison between δ_0 and δ_2 is ambiguous. When θ is near $1/2$, δ_2 is the preferable estimator, but if θ is near 0 or 1, δ_0 is preferable. If θ were known,

we could choose between δ_0 and δ_2 . However, if θ were known, there would be no need to estimate its value.

3.2 Sufficient Statistics

Suppose X and Y are independent with common Lebesgue density

$$f_\theta(x) = \begin{cases} \theta e^{-\theta x}, & x \geq 0; \\ 0, & x < 0, \end{cases}$$

and let U be independent of X and Y and uniformly distributed on $(0, 1)$. Take $T = X + Y$, and define

$$\tilde{X} = UT \text{ and } \tilde{Y} = (1 - U)T.$$

Let us find the joint density of $\tilde{X}$ and $\tilde{Y}$. The density of T is needed, and this can be found by smoothing. Because X and Y are independent,¹

$$\begin{aligned} P(T \leq t | Y = y) &= P(X + Y \leq t | Y = y) \\ &= E[I\{X + Y \leq t\} | Y = y] \\ &= \int I\{x + y \leq t\} dP_X(x) \\ &= F_X(t - y). \end{aligned}$$

So $P(T \leq t | Y) = F_X(t - Y)$ and

$$F_T(t) = P(T \leq t) = E F_X(t - Y).$$

This formula holds generally. Specializing to our specific problem, $F_X(t - Y)$ is $1 - e^{-\theta(t-Y)}$ on $Y < t$ and is zero on $Y \geq t$. Writing the expected value of this variable as an integral against the density of Y , for $t \geq 0$,

$$F_T(t) = \int_0^t (1 - e^{-\theta(t-y)}) \theta e^{-\theta y} dy = 1 - e^{-\theta t} - t\theta e^{-\theta t}.$$

Taking derivatives, T has density

$$p_T(t) = F'_T(t) = t\theta^2 e^{-\theta t}, \quad t \geq 0,$$

with $p_T(t) = 0$ for $t < 0$. Because T and U are independent, they have joint density

$$p_\theta(t, u) = \begin{cases} t\theta^2 e^{-\theta t}, & t \geq 0, u \in (0, 1); \\ 0, & \text{otherwise.} \end{cases}$$

¹ The distribution of T naturally depends on θ , but for convenience this dependence is suppressed in the notation here.

From this,

$$P\left(\begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix} \in B\right) = \iint 1_B(tu, t(1-u)) p_\theta(t, u) du dt.$$

Changing variables to $x = ut$, $du = dx/t$ in the inner integral, and reversing the order of integration using Fubini's theorem,

$$P\left(\begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix} \in B\right) = \iint 1_B(x, t-x) t^{-1} p_\theta(t, x/t) dt dx.$$

Now a change of variables to $y = t - x$ in the inner integral gives

$$P\left(\begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix} \in B\right) = \iint 1_B(x, y) (x+y)^{-1} p_\theta\left(x+y, \frac{x}{x+y}\right) dy dx.$$

Thus $\tilde{X}$ and $\tilde{Y}$ have joint density

$$\frac{p_\theta\left(x+y, \frac{x}{x+y}\right)}{x+y} = \begin{cases} \theta^2 e^{-\theta(x+y)}, & x \geq 0, y \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

This density is the same as the joint density of X and Y , and so this calculation shows that the joint distribution of $\tilde{X}$ and $\tilde{Y}$ is the same as the joint distribution of X and Y . Considered as data that provide information about θ , the pair $(\tilde{X}, \tilde{Y})$ should be just as informative as (X, Y) . But $(\tilde{X}, \tilde{Y})$ can be computed from $T = X + Y$ and U . Because the distribution of U does not depend on θ , it could be generated numerically on a computer or obtained from a table of random numbers. Thus T by itself also provides as much information about θ as the pair (X, Y) because we could construct fake data $(\tilde{X}, \tilde{Y})$ equivalent to (X, Y) using any convenient variable U that is uniformly distributed on $(0, 1)$. The sum $T = X + Y$ is called a *sufficient statistic*. This construction of fake data works because the conditional distribution Q_t for X and Y given $T = t$, given explicitly by²

$$Q_t(B) = P_\theta\left[\begin{pmatrix} X \\ Y \end{pmatrix} \in B \mid T = t\right] = P\left[\begin{pmatrix} Ut \\ (1-U)t \end{pmatrix} \in B\right],$$

does not depend on θ . This motivates the following definition in a general setting.

Definition 3.2. Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. Then $T = T(X)$ is a sufficient statistic for $\mathcal{P}$ (or for X , or for θ) if for every t and θ , the conditional distribution of X under P_θ given $T = t$ does not depend on θ .

² See Chapter 6 for a proper treatment of conditional distributions.

Suppose T is sufficient, and let

$$Q_t(B) = P_\theta(X \in B|T = t).$$

Then $P_\theta(X \in B|T) = Q_T(B)$ and, by smoothing,

$$P_\theta(X \in B) = E_\theta P_\theta(X \in B|T) = E_\theta Q_T(B).$$

Suppose we use a random number generator to construct “fake” data $\tilde{X}$ from T taking $\tilde{X} \sim Q_t$ when $T = t$. Then

$$\tilde{X}|T = t \sim Q_t$$

and by smoothing

$$P_\theta(\tilde{X} \in B) = E_\theta P_\theta(\tilde{X} \in B|T) = E_\theta Q_T(B). \quad (3.1)$$

So X and $\tilde{X}$ have the same distribution.

Theorem 3.3. *Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ and that $T = T(X)$ is sufficient. Then for any estimator $\delta(X)$ of $g(\theta)$ there exists a randomized estimator based on T that has the same risk function as $\delta(X)$.*

Proof. A randomized estimator is one that can be constructed from T with auxiliary random number generation. Inasmuch as $\tilde{X}$ can be constructed from T by random number generation, $\delta(\tilde{X})$ is a randomized estimator, and its risk is the same as the risk of $\delta(X)$ because X and $\tilde{X}$ both have the same distribution P_θ . $\square$

A similar notion of sufficiency, due to Blackwell (1951), can be used to compare experiments. It is natural here to identify an experiment with the model that gives distributions for data from the experiment. Let $\tilde{\mathcal{P}} = \{\tilde{P}_\theta : \theta \in \Omega\}$ and $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be models for two experiments. As before, the notion is that $\tilde{\mathcal{P}}$ is sufficient if fake data can be created using an observation from $\tilde{\mathcal{P}}$ and external randomization, with the distributions for the fake data, as θ varies, the same as distributions for real data from the other experiment.

Definition 3.4. *Model $\tilde{\mathcal{P}}$ is sufficient for $\mathcal{P}$ if there is a stochastic transition kernel Q such that*

$$P_\theta(B) = \int Q_t(B) d\tilde{P}_\theta(t)$$

for every Borel set B and all $\theta \in \Omega$.

The argument that this supports randomization to construct fake data is the same as (3.1). Suppose $\tilde{\mathcal{P}}$ is sufficient for $\mathcal{P}$, and let $T \sim \tilde{P}_\theta$ be data from the sufficient experiment. When $T = t$, the fake data $\tilde{X}$ is obtained by sampling from Q_t , so that

$$P(\tilde{X} \in B | T = t) = Q_t(B).$$

Then, by smoothing,

$$P(\tilde{X} \in B) = EP(\tilde{X} \in B | T) = EQ_T(B) = \int Q_t(B) d\tilde{P}_\theta(t) = P_\theta(B).$$

This shows that $\tilde{X} \sim P_\theta$, so the distributions for $\tilde{X}$, as desired, are the same as distributions for real data from the other experiment $\mathcal{P}$, regardless of the value of $\theta \in \Omega$.

3.3 Factorization Theorem

From the definition of sufficiency in the last section it is easy to understand the sense in which a sufficient statistic $T(X)$ conveys all of the information about θ from data X , at least when the model is correct. But the definition is less useful for finding a sufficient statistic, or trying to determine whether a specific statistic is sufficient. When distributions in the family are specified by densities, sufficiency can be checked using the factorization theorem simply by looking at the form of the densities.

Definition 3.5. A family of distributions $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is dominated if there exists a measure μ with P_θ absolutely continuous with respect to μ , for all $\theta \in \Omega$.

Theorem 3.6 (Factorization Theorem). Let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a family of distributions dominated by μ . A necessary and sufficient condition for a statistic T to be sufficient is that there exist functions $g_\theta \geq 0$ and $h \geq 0$ such that the densities p_θ for the family satisfy

$$p_\theta(x) = g_\theta(T(x))h(x), \text{ for a.e. } x \text{ under } \mu.$$

A proof of this result is given in Section 6.4. It depends in part on a proper definition of conditional distributions.

Example 3.7. Suppose $X_1, \dots, X_n$ are i.i.d. absolutely continuous variables with common marginal density

$$f_\theta(x) = \begin{cases} (\theta + 1)x^\theta, & x \in (0, 1); \\ 0, & \text{otherwise,} \end{cases}$$

for $\theta > -1$. Then their joint density p_θ is

$$p_\theta(x) = \prod_{i=1}^n f_\theta(x_i) = \prod_{i=1}^n (\theta + 1)x_i^\theta = (\theta + 1)^n \left(\prod_{i=1}^n x_i \right)^\theta, \quad x \in (0, 1)^n,$$

with $p_\theta(x) = 0$ if $x \notin (0, 1)^n$. Taking $g_\theta(t) = (\theta + 1)^n t^\theta$ and $h = 1_{(0,1)^n}$, from the factorization theorem, $T = \prod_{i=1}^n X_i$ is sufficient.

When $\{x : p_\theta(x) > 0\}$ depends on θ , care is needed to ensure that formulas for p_θ used in the factorization theorem work for all x . To accomplish this, indicator functions are often used.

Example 3.8. Suppose $X_1, \dots, X_n$ are a random sample from the uniform distribution on $(\theta, \theta + 1)$. Then the common marginal density is

$$f_\theta(x) = \begin{cases} 1, & x \in (\theta, \theta + 1); \\ 0, & \text{otherwise.} \end{cases}$$

So $f_\theta = 1_{(\theta, \theta+1)}$. The joint density is

$$p_\theta(x) = \prod_{i=1}^n 1_{(\theta, \theta+1)}(x_i),$$

which equals one if and only if $\max_i x_i < \theta + 1$ and $\min_i x_i > \theta$. So

$$p_\theta(x) = 1_{(\theta, \infty)}(\min_i x_i) 1_{(-\infty, \theta+1)}(\max_i x_i).$$

By the factorization theorem, $T = (\min_i X_i, \max_i X_i)$ is sufficient.

3.4 Minimal Sufficiency

If T is sufficient for a family of distributions $\mathcal{P}$, and if $T = f(\tilde{T})$, then $\tilde{T}$ is also sufficient. This follows easily from the factorization theorem when the family is dominated. This suggests the following definition.

Definition 3.9. A statistic T is minimal sufficient if T is sufficient, and for every sufficient statistic $\tilde{T}$ there exists a function f such that $T = f(\tilde{T})$ (a.e. $\mathcal{P}$). Here (a.e. $\mathcal{P}$) means that the set where equality fails is a null set for every $P \in \mathcal{P}$.

Example 3.10. If $X_1, \dots, X_{2n}$ are i.i.d. from $N(\theta, 1)$, $\theta \in \mathbb{R}$, then

$$\tilde{T} = \left(\begin{array}{c} \sum_{i=1}^n X_i \\ \sum_{i=n+1}^{2n} X_i \end{array} \right)$$

is sufficient but not minimal. It can be shown that $T = \sum_{i=1}^{2n} X_i$ is minimal sufficient here,³ and $T = f(\tilde{T})$ if we take $f(t) = t_1 + t_2$.

If $\mathcal{P}$ is a dominated family, then the density $p_\theta(X)$, viewed as a function of θ , is called the *likelihood function*. By the factorization theorem, any sufficient statistic must provide enough information to graph the shape of the likelihood, where two functions are defined to have the same shape if they are proportional. The next result shows that a statistic is minimal sufficient if there is a one-to-one relation between the statistic and the likelihood shape.

³ This follows from Example 3.12 below.

Theorem 3.11. Suppose $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is a dominated family with densities $p_\theta(x) = g_\theta(T(x))h(x)$. If $p_\theta(x) \propto_\theta p_\theta(y)$ implies $T(x) = T(y)$, then T is minimal sufficient.⁴

Proof. A proper proof of this result unfortunately involves measure-theoretic niceties, but here is the basic idea. Suppose $\tilde{T}$ is sufficient. Then $p_\theta(x) = \tilde{g}_\theta(\tilde{T}(x))\tilde{h}(x)$ (a.e. μ). Assume this equation holds for all x . If T is not a function of $\tilde{T}$, then there must be two data sets x and y that give the same value for $\tilde{T}$, $\tilde{T}(x) = \tilde{T}(y)$, but different values for T , $T(x) \neq T(y)$. But then

$$p_\theta(x) = \tilde{g}_\theta(\tilde{T}(x))\tilde{h}(x) \propto_\theta \tilde{g}_\theta(\tilde{T}(y))\tilde{h}(y) = p_\theta(y),$$

and from the condition on T in the theorem, $T(x)$ must equal $T(y)$. Thus T is a function of $\tilde{T}$. Because $\tilde{T}$ was an arbitrary sufficient statistic, T is minimal. $\square$

Although a proper development takes more work, this result in essence says that the shape of the likelihood is minimal sufficient, and so a minimal sufficient “statistic” exists for dominated families.⁵ When this result is used, if the implication only fails on a null set (for the family), T will still be minimal sufficient. In particular, if the implication holds unless $p_\theta(x)$ and $p_\theta(y)$ are identically zero as θ varies, then T will be minimal sufficient.

Example 3.12. Suppose $\mathcal{P}$ is an s -parameter exponential family with densities

$$p_\theta(x) = e^{\eta(\theta) \cdot T(x) - B(\theta)} h(x),$$

for $\theta \in \Omega$. By the factorization theorem, T is sufficient. Suppose $p_\theta(x) \propto_\theta p_\theta(y)$. Then

$$e^{\eta(\theta) \cdot T(x)} \propto_\theta e^{\eta(\theta) \cdot T(y)},$$

which implies that

$$\eta(\theta) \cdot T(x) = \eta(\theta) \cdot T(y) + c,$$

where the constant c may depend on x and y , but is independent of θ . If θ_0 and θ_1 are any two points in Ω ,

$$[\eta(\theta_0) - \eta(\theta_1)] \cdot T(x) = [\eta(\theta_0) - \eta(\theta_1)] \cdot T(y)$$

and

⁴ The notation “ $\propto_\theta$ ” here means that the two expressions are proportional when viewed as functions of θ . So $p_\theta(x) \propto_\theta p_\theta(y)$ here would mean that there is a “proportionality constant” c that may depend on x and y , so $c = c(x, y)$, such that $p_\theta(x) = c(x, y)p_\theta(y)$, for all $\theta \in \Omega$.

⁵ At a technical level this may fail without a bit of regularity. Minimal sufficient σ -fields must exist in this setting, but there may be no minimal sufficient statistic if $\mathcal{P}$ is not separable (under total variation norm). For discussion and counterexamples, see Bahadur (1954) and Landers and Rogge (1972).

$$[\eta(\theta_0) - \eta(\theta_1)] \cdot [T(x) - T(y)] = 0.$$

This shows that $T(x) - T(y)$ is orthogonal to every vector in

$$\eta(\Omega) \ominus \eta(\Omega) \stackrel{\text{def}}{=} \{\eta(\theta_0) - \eta(\theta_1) : \theta_0 \in \Omega, \theta_1 \in \Omega\},$$

and so it must lie in the orthogonal complement of the linear span⁶ of $\eta(\Omega) \ominus \eta(\Omega)$. In particular, if the linear span of $\eta(\Omega) \ominus \eta(\Omega)$ is all of $\mathbb{R}^s$, then $T(x)$ must equal $T(y)$. So, in this case, T will be minimal sufficient.

Example 3.13. Suppose $X_1, \dots, X_n$ are i.i.d. absolutely continuous variables with common marginal density

$$f_\theta(x) = \frac{1}{2}e^{-|x-\theta|}.$$

Then the joint density is

$$p_\theta(x) = \frac{1}{2^n} \exp \left\{ - \sum_{i=1}^n |x_i - \theta| \right\}.$$

The variables $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ found by listing $X_1, \dots, X_n$ in increasing order are called the *order statistics*. By the factorization theorem, $T = (X_{(1)}, \dots, X_{(n)})'$, is sufficient. Suppose $p_\theta(x) \propto_\theta p_\theta(y)$. Then the difference between $\sum_{i=1}^n |x_i - \theta|$ and $\sum_{i=1}^n |y_i - \theta|$ is constant in θ . Both of these functions are piecewise linear functions of θ with a slope that increases by two at each order statistic. The difference can only be constant in θ if x and y have the same order statistics. Thus the order statistics are minimal sufficient for this family of distributions.

3.5 Completeness

Completeness is a technical condition that strengthens sufficiency in a useful fashion.

Definition 3.14. A statistic T is complete for a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ if

$$E_\theta f(T) = c, \text{ for all } \theta,$$

implies $f(T) = c$ (a.e. $\mathcal{P}$).

Remark 3.15. Replacing f by $f - c$, the constant c in this definition could be taken to be zero.

⁶ See Appendix A.3 for a review of vector spaces and the geometry of $\mathbb{R}^n$.

Example 3.16. Suppose $X_1, \dots, X_n$ are i.i.d. from a uniform distribution on $(0, \theta)$. Using indicator functions, the joint density is $I\{\min x_i > 0\}I\{\max x_i < \theta\}/\theta^n$, and so $T = \max\{X_1, \dots, X_n\}$ is sufficient by the factorization theorem (Theorem 3.6). By independence, for $t \in (0, \theta)$,

$$\begin{aligned} P_\theta(T \leq t) &= P_\theta(X_1 \leq t, \dots, X_n \leq t) \\ &= P_\theta(X_1 \leq t) \times \dots \times P_\theta(X_n \leq t) = (t/\theta)^n. \end{aligned}$$

Differentiating this expression, T has density nt^{n-1}/θ^n , $t \in (0, \theta)$. Suppose $E_\theta f(T) = c$ for all $\theta > 0$; then

$$E_\theta[f(T) - c] = \frac{n}{\theta^n} \int_0^\theta [f(t) - c]t^{n-1} dt = 0.$$

From this (using fact 4 about integration in Section 1.4) $[f(t) - c]t^{n-1} = 0$ for a.e. $t > 0$. So $f(T) = c$ (a.e. $\mathcal{P}$), and T is complete.

Theorem 3.17. *If T is complete and sufficient, then T is minimal sufficient.*

Proof. Let $\tilde{T}$ be a minimal sufficient statistic, and assume T and $\tilde{T}$ are both bounded random variables. Then $\tilde{T} = f(T)$. Define $g(\tilde{T}) = E_\theta[T|\tilde{T}]$, noting that this function is independent of θ because $\tilde{T}$ is sufficient. By smoothing, $E_\theta g(\tilde{T}) = E_\theta T$, and so $E_\theta[T - g(\tilde{T})] = 0$, for all θ . But $T - g(\tilde{T}) = T - g(f(T))$, a function of T , and so by completeness, $T = g(\tilde{T})$ (a.e. $\mathcal{P}$). This establishes a one-to-one relationship between T and $\tilde{T}$. From the definition of minimal sufficiency, T must also be minimal sufficient.

For the general case, first note that sufficiency and completeness are both preserved by one-to-one transformations, so two statistics can be considered equivalent if they are related by a one-to-one (bimeasurable) function. But there are one-to-one bimeasurable functions from $\mathbb{R}^n$ to $\mathbb{R}$, and so any random vector is equivalent to a single random variable.⁷ Using this, if T and $\tilde{T}$ are random vectors, the result follows easily from the one-dimensional case transforming both of them to equivalent random variables. $\square$

Definition 3.18. *An exponential family with densities $p_\theta(x) = \exp\{\eta(\theta) \cdot T(x) - B(\theta)\}h(x)$, $\theta \in \Omega$, is said to be of full rank if the interior of $\eta(\Omega)$ is not empty and if $T_1, \dots, T_s$ do not satisfy a linear constraint of the form $v \cdot T = c$ (a.e. μ).*

If $\Omega \subset \mathbb{R}^s$ and η is continuous and one-to-one (injective), and the interior of Ω is nonempty, then the interior of $\eta(\Omega)$ cannot be empty. This follows from the “invariance of domain” theorem of Brouwer (1912).

If the interior of $\eta(\Omega)$ is not empty, then the linear span of $\eta(\Omega) \ominus \eta(\Omega)$ will be all of $\mathbb{R}^s$, and, by Example 3.12, T will be minimal sufficient. The following result shows that in this case T is also complete.

⁷ For instance, the function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ that alternates the decimal digits of its arguments, and thus, for instance, $g(12.34\dots, 567.89\dots) = 506172.8394\dots$, is one-to-one and bimeasurable.

Theorem 3.19. *In an exponential family of full rank, T is complete.*

Definition 3.20. *A statistic V is called ancillary if its distribution does not depend on θ . So, V by itself provides no information about θ .*

An ancillary statistic V , by itself, provides no useful information about θ . But in some situations V can be a function of a minimal sufficient statistic T . For instance, in Example 3.13 differences $X_{(i)} - X_{(j)}$ between order statistics are ancillary. But they are functions of the minimal sufficient T and are relevant to inference. The following result of Basu shows that when T is complete it will contain no ancillary information. See Basu (1955, 1958) or Lehmann (1981) for further discussion.

Theorem 3.21 (Basu). *If T is complete and sufficient for $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$, and if V is ancillary, then T and V are independent under P_θ for any $\theta \in \Omega$.*

Proof. Define $q_A(t) = P_\theta(V \in A | T = t)$, so that $q_A(T) = P_\theta(V \in A | T)$, and define $p_A = P_\theta(V \in A)$. By sufficiency and ancillarity, neither p_A nor $q_A(t)$ depend on θ . Also, by smoothing,

$$p_A = P_\theta(V \in A) = E_\theta P_\theta(V \in A | T) = E_\theta q_A(T),$$

and so, by completeness, $q_A(T) = p_A$ (a.e. $\mathcal{P}$). By smoothing,

$$\begin{aligned} P_\theta(T \in B, V \in A) &= E_\theta 1_B(T) 1_A(V) \\ &= E_\theta E_\theta(1_B(T) 1_A(V) \mid T) \\ &= E_\theta 1_B(T) E_\theta(1_A(V) \mid T) \\ &= E_\theta 1_B(T) q_A(T) \\ &= E_\theta 1_B(T) p_A \\ &= P_\theta(T \in B) P_\theta(V \in A). \end{aligned}$$

Here A and B are arbitrary Borel sets, and so T and V are independent. $\square$

Example 3.22. Suppose $X_1, \dots, X_n$ are i.i.d. from $N(\mu, \sigma^2)$, and take $\mathcal{P} = \mathcal{P}_\sigma = \{N(\mu, \sigma^2)^n : \mu \in \mathbb{R}\}$. (Thus $\mathcal{P}_\sigma$ is the family of all normal distributions with standard deviation the fixed value σ .) With $\bar{x} = (x_1 + \dots + x_n)/n$, the joint density can be written as

$$\frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[\frac{n\mu}{\sigma^2} \bar{x} - \frac{n\mu^2}{2\sigma^2} - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 \right].$$

These densities for $\mathcal{P}_\sigma$ form a full rank exponential family, and so the average $\bar{X} = (X_1 + \dots + X_n)/n$ is a complete sufficient statistic for $\mathcal{P}_\sigma$. Define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

called the sample variance. For the family $\mathcal{P}_\sigma$, S^2 is ancillary. To see this, let $Y_i = X_i - \mu$, $i = 1, \dots, n$. Because

$$\begin{aligned} P_\mu(Y_i \leq y) &= P_\mu(X_i \leq y + \mu) = \int_{-\infty}^{y+\mu} \exp\left[-\frac{1}{2\sigma^2}(x - \mu)^2\right] \frac{dx}{\sqrt{2\pi\sigma^2}} \\ &= \int_{-\infty}^y \exp\left[-\frac{1}{2\sigma^2}u^2\right] \frac{du}{\sqrt{2\pi\sigma^2}}, \end{aligned}$$

and the integrand is the density for $N(0, \sigma^2)$, $Y_i \sim N(0, \sigma^2)$. Then $Y_1, \dots, Y_n$ are i.i.d. from $N(0, \sigma^2)$. Because $\bar{Y} = (Y_1 + \dots + Y_n)/n = \bar{X} - \mu$, $X_i - \bar{X} = Y_i - \bar{Y}$, $i = 1, \dots, n$, and

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

Because the joint distribution of $Y_1, \dots, Y_n$ depends on σ but not μ , S^2 is ancillary for $\mathcal{P}_\sigma$. Hence, by Basu's theorem, $\bar{X}$ and S^2 are independent.⁸

3.6 Convex Loss and the Rao–Blackwell Theorem

Definition 3.23. A real-valued function f on a convex set $\mathcal{C}$ in $\mathbb{R}^p$ is called convex if, for any $x \neq y$ in $\mathcal{C}$ and any $\gamma \in (0, 1)$,

$$f[\gamma x + (1 - \gamma)y] \leq \gamma f(x) + (1 - \gamma)f(y). \quad (3.2)$$

The function f is strictly convex if (3.2) holds with strict inequality.

Geometrically, f is strictly convex if the graph of f for values between x and y lies below the chord joining $(x, f(x))$ and $(y, f(y))$, illustrated in Figure 3.3. If $p = 1$ and f'' exists and is nonnegative on $\mathcal{C}$, the f is convex. The next result is the supporting hyperplane theorem in one dimension.

Theorem 3.24. If f is a convex function on an open interval $\mathcal{C}$, and if t is an arbitrary point in $\mathcal{C}$, then there exists a constant $c = c_t$ such that

$$f(t) + c(x - t) \leq f(x), \quad \forall x \in \mathcal{C}.$$

If f is strictly convex, then $f(t) + c(x - t) < f(x)$ for all $x \in \mathcal{C}$, $x \neq t$.

The left-hand side of this inequality is a line through $(t, f(t))$. So, this result says that we can always find a line below the graph of f touching the graph of f at t . This is illustrated in Figure 3.4.

⁸ The independence established here plays an important role when distribution theory for this example is considered in more detail in Section 4.3. Independence can also be established using spherical symmetry of the multivariate normal distribution, an approach developed in a more general setting in Chapter 14.

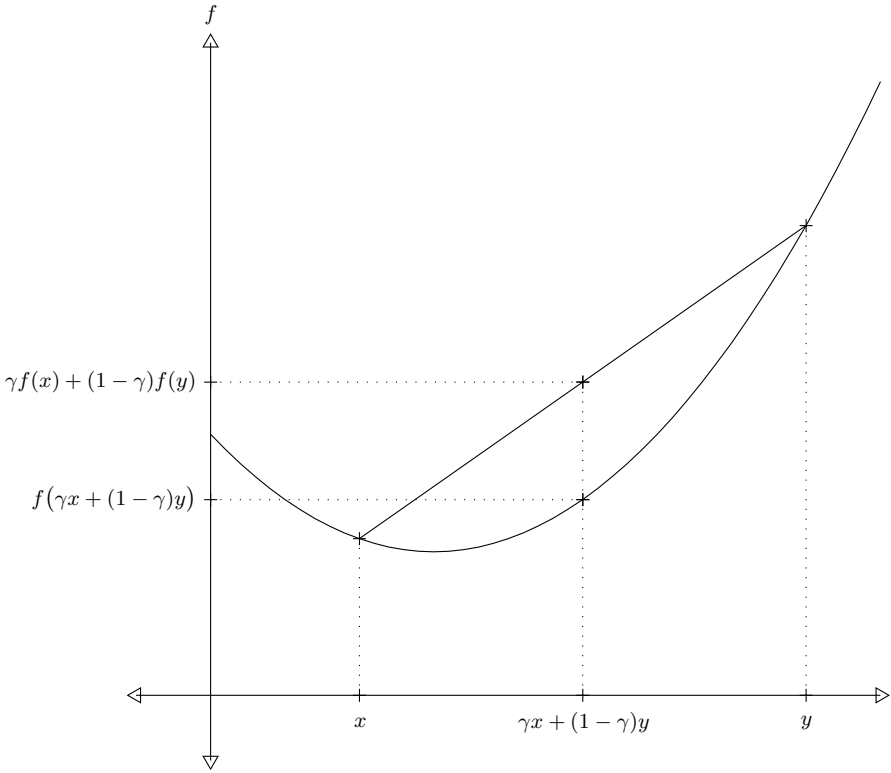


Fig. 3.3. A convex function.

Theorem 3.25 (Jensen's Inequality). *If $\mathcal{C}$ is an open interval, f is a convex function on $\mathcal{C}$, $P(X \in \mathcal{C}) = 1$, and EX is finite, then*

$$f(EX) \leq Ef(X).$$

If f is strictly convex, the inequality is strict unless X is almost surely constant.

Proof. By Theorem 3.24 with $t = EX$, for some constant c ,

$$f(EX) + c(x - EX) \leq f(x), \quad \forall x \in \mathcal{C},$$

and so

$$f(EX) + c(X - EX) \leq f(X), \quad (\text{a.e. } P).$$

The first assertion of the theorem follows taking expectations. If f is strictly convex this bound will be strict on $X \neq EX$. The second assertion of the theorem then follows using fact 2 from Section 1.4. $\square$

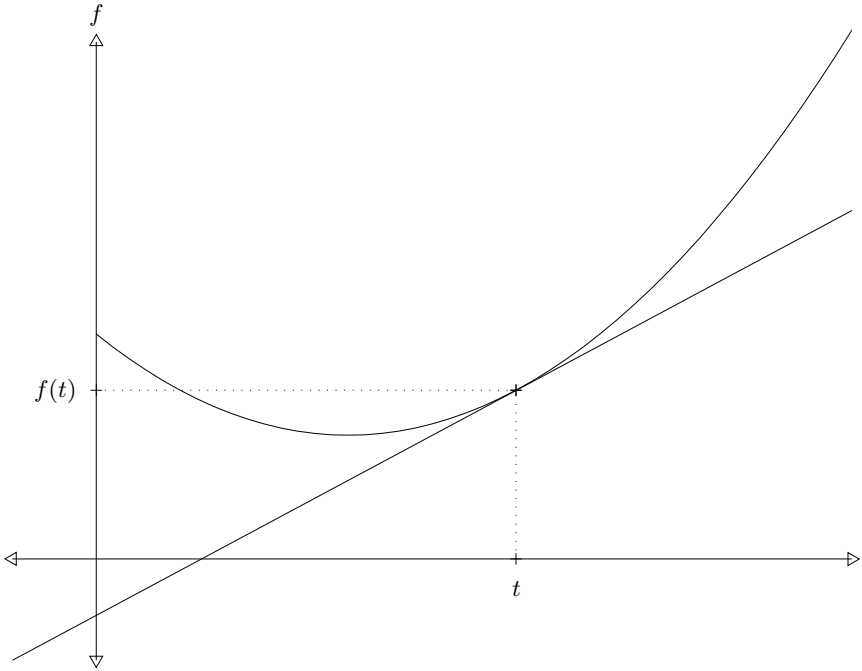


Fig. 3.4. Convex function with support line.

Remark 3.26. Jensen's inequality also holds in higher dimensions, with X a random vector.

Example 3.27. The functions $1/x$ and $-\log x$ are strictly convex on $(0, \infty)$. If $X > 0$, then $1/EX \leq E[1/X]$ and $\log EX \geq E \log(X)$. These inequalities are strict unless X is constant.

If $\delta(X)$ is an estimator of $g(\theta)$, then the risk of δ for a loss function $L(\theta, d)$ is $R(\delta, \theta) = E_{\theta} L(\theta, \delta(X))$. Suppose T is a sufficient statistic. By Theorem 3.3 there is a randomized estimator based on T with the same risk as δ . The following result shows that for convex loss functions there is generally a non-randomized estimator based on T that has smaller risk than δ .

Theorem 3.28 (Rao–Blackwell). *Let T be a sufficient statistic for $\mathcal{P} = \{P_{\theta} : \theta \in \Omega\}$, let δ be an estimator of $g(\theta)$, and define $\eta(T) = E[\delta(X)|T]$. If $\theta \in \Omega$, $R(\theta, \delta) < \infty$, and $L(\theta, \cdot)$ is convex, then*

$$R(\theta, \eta) \leq R(\theta, \delta).$$

Furthermore, if $L(\theta, \cdot)$ is strictly convex, the inequality will be strict unless $\delta(X) = \eta(T)$ (a.e. P_{θ}).

Proof. Jensen's inequality with expectations against the conditional distribution of $\delta(X)$ given T gives

$$L(\theta, \eta(T)) \leq E_{\theta}[L(\theta, \delta(X)) \mid T].$$

Taking expectations, $R(\theta, \eta) \leq R(\theta, \delta)$. The assertion about strict inequality follows after a bit of work from the second assertion in Jensen's inequality. $\square$

This result shows that with convex loss functions the only estimators worth considering, at least if estimators are judged solely by their risk, are functions of T but not X . It can also be used to show that any randomized estimator is worse than a corresponding nonrandomized estimator. Using the probability integral transformation,⁹ any randomized estimator can be viewed as a function of X and U , where X and U are independent and the distribution of U does not depend on θ . But if X and U are both considered as data, then X is sufficient, and with convex loss the risk of a randomized estimator $\delta(X, U)$ will be worse than the risk of the estimator $E(\delta(X, U) \mid X)$, which is based solely on X .

3.7 Problems¹⁰

1. An estimator δ is called *inadmissible* if there is a competing estimator $\tilde{\delta}$ with a better risk function, that is, if $R(\theta, \tilde{\delta}^*) \leq R(\theta, \delta)$, for all $\theta \in \Omega$, and $R(\theta, \tilde{\delta}^*) < R(\theta, \delta)$, for some $\theta \in \Omega$. If there is no estimator with a better risk function, δ is *admissible*. Consider estimating success probability $\theta \in [0, 1]$ from data $X \sim \text{Binomial}(n, \theta)$ under squared error loss. Define $\delta_{a,b}$ by

$$\delta_{a,b}(X) = a \frac{X}{n} + (1-a)b,$$

which might be called a linear estimator, because it is a linear function of X .

- a) Find the variance and bias of $\delta_{a,b}$. (The bias of an arbitrary estimator δ of θ is defined as $b(\theta, \delta) = E_{\theta}\delta(X) - \theta$.)
- b) If $a > 1$, show that $\delta_{a,b}$ is inadmissible by finding a competing linear estimator with better risk. Hint: The risk of an arbitrary estimator δ under squared error loss is $\text{Var}_{\theta}(\delta(X)) + b^2(\theta, \delta)$. Find an unbiased estimator with smaller variance.
- c) If $b > 1$ or $b < 0$, and $a \in [0, 1]$, show that $\delta_{a,b}$ is inadmissible by finding a competing linear estimator with better risk. Hint: Find an estimator with the same variance but better bias.

⁹ If the inverse for a possibly discontinuous cumulative distribution function F is defined as $F^{\leftarrow}(t) = \sup\{x : F(x) \leq t\}$, and if U is uniformly distributed on $(0, 1)$, then the random variable $F^{\leftarrow}(U)$ has cumulative distribution function F .

¹⁰ Solutions to the starred problems are given at the back of the book.

d) If $a < 0$, find a linear estimator with better risk than $\delta_{a,b}$.

*2. Suppose data $X_1, \dots, X_n$ are independent with

$$P_\theta(X_i \leq x) = x^{t_i \theta}, \quad x \in (0, 1),$$

where $\theta > 0$ is the unknown parameter, and $t_1, \dots, t_n$ are known positive constants. Find a one-dimensional sufficient statistic T .

*3. An object with weight θ is weighed on scales with different precision. The data $X_1, \dots, X_n$ are independent, with $X_i \sim N(\theta, \sigma_i^2)$, $i = 1, \dots, n$, with the standard deviations $\sigma_1, \dots, \sigma_n$ known constants. Use sufficiency to suggest a weighted average of $X_1, \dots, X_n$ to estimate θ . (A weighted average would have form $\sum_{i=1}^n w_i X_i$, where the w_i are positive and sum to one.)

*4. Let $X_1, \dots, X_n$ be a random sample from an arbitrary discrete distribution P on $\{1, 2, 3\}$. Find a two-dimensional sufficient statistic.

5. For $\theta \in \Omega = (0, 1)$, let $\tilde{P}_\theta$ denote a discrete distribution with mass function

$$\tilde{p}_\theta(t) = (1+t)\theta^2(1-\theta)^t, \quad t = 0, 1, \dots,$$

and let P_θ denote the binomial distribution with two trials and success probability θ . Show that the model $\tilde{\mathcal{P}} = \{\tilde{P}_\theta : \theta \in \Omega\}$ is sufficient for the binomial model $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. Identify the stochastic transition Q by giving the mass functions

$$q_t(x) = Q_t(\{x\}), \quad x = 0, 1, 2,$$

for $t = 0, 1, \dots$

*6. The beta distribution with parameters $\alpha > 0$ and $\beta > 0$ has density

$$f_{\alpha,\beta}(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, & x \in (0, 1); \\ 0, & \text{otherwise.} \end{cases}$$

Suppose $X_1, \dots, X_n$ are i.i.d. from a beta distribution.

a) Determine a minimal sufficient statistic (for the family of joint distributions) if α and β vary freely.

b) Determine a minimal sufficient statistic if $\alpha = 2\beta$.

c) Determine a minimal sufficient statistic if $\alpha = \beta^2$.

*7. *Logistic regression.* Let $X_1, \dots, X_n$ be independent Bernoulli variables, with $p_i = P(X_i = 1)$, $i = 1, \dots, n$. Let $t_1, \dots, t_n$ be a sequence of known constants that are related to the p_i via

$$\log \frac{p_i}{1-p_i} = \alpha + \beta t_i,$$

where α and β are unknown parameters. Determine a minimal sufficient statistic for the family of joint distributions.

- *8. The multinomial distribution, derived later in Section 5.3, is a discrete distribution with mass function

$$\frac{n!}{x_1! \times \cdots \times x_s!} p_1^{x_1} \times \cdots \times p_s^{x_s},$$

where $x_0, \dots, x_s$ are nonnegative integers summing to n , where $p_1, \dots, p_s$ are nonnegative probabilities summing to one, and n is the sample size. Let $N_{11}, N_{12}, N_{21}, N_{22}$ have a multinomial distribution with n trials and success probabilities $p_{11}, p_{12}, p_{21}, p_{22}$. (A common model for a two-by-two contingency table.)

- a) Give a minimal sufficient statistic if the success probabilities vary freely over the unit simplex in $\mathbb{R}^4$. (The unit simplex in $\mathbb{R}^p$ is the set of all vectors with nonnegative entries summing to one.)
- b) Give a minimal sufficient statistic if the success probabilities are constrained so that $p_{11}p_{22} = p_{12}p_{21}$.

- *9. Let f be a positive integrable function on $(0, \infty)$. Define

$$c(\theta) = 1 / \int_{\theta}^{\infty} f(x) dx,$$

and take $p_{\theta}(x) = c(\theta)f(x)$ for $x > \theta$, and $p_{\theta}(x) = 0$ for $x \leq \theta$. Let $X_1, \dots, X_n$ be i.i.d. with common density p_{θ} .

- a) Show that $M = \min\{X_1, \dots, X_n\}$ is sufficient.
- b) Show that M is minimal sufficient.

- *10. Suppose $X_1, \dots, X_n$ are i.i.d. with common density $f_{\theta}(x) = (1 + \theta x)/2$, $|x| < 1$; $f_{\theta}(x) = 0$, otherwise, where $\theta \in [-1, 1]$ is an unknown parameter. Show that the order statistics are minimal sufficient. (Hint: A polynomial of degree n is uniquely determined by its value on a grid of $n + 1$ points.)
11. Consider a two-sample problem in which $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma_x^2)$ and $Y_1, \dots, Y_m$ is an independent random sample from $N(\mu, \sigma_y^2)$. Let P_{θ} denote the joint distribution of these $n + m$ variables, with $\theta = (\mu, \sigma_x^2, \sigma_y^2)$. Find a minimal sufficient statistic for this family of distributions.
12. Let Z_1 and Z_2 be independent standard normal random variables with common density $\phi(x) = \exp(-x^2/2)/\sqrt{2\pi}$, and suppose X and Y are related to these variables by

$$X = Z_1 \text{ and } Y = (X + Z_2)\theta,$$

where $\theta > 0$ is an unknown parameter. (This might be viewed as a regression model in which the independent variable is measured with error.)

- a) Find the joint density for X and Y .
- b) Suppose our data $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. random vectors with common distribution that of X and Y in part (a),

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \begin{pmatrix} X \\ Y \end{pmatrix}, \quad i = 1, \dots, n.$$

Find a minimal sufficient statistic.

13. Let $X_1, \dots, X_n$ be independent Poisson variables with $\lambda_i = EX_i$, $i = 1, \dots, n$. Let $t_1, \dots, t_n$ be a sequence of known constants related to the λ_i by

$$\log \lambda_i = \alpha + \beta t_i, \quad i = 1, \dots, n,$$

where α and β are unknown parameters. Find a minimal sufficient statistic for the family of joint distributions.

14. Let $X_1, \dots, X_n$ be i.i.d. from a discrete distribution Q on $\{1, 2, 3\}$. Let $p_i = Q(\{i\}) = P(X_j = i)$, $i = 1, 2, 3$, and assume we know that $p_1 = 1/3$, but have no additional knowledge of Q . Define

$$N_i = \#\{j \leq n : X_j = i\}.$$

- a) Show that $T = (N_1, N_2)$ is sufficient.
 b) Is T minimal sufficient? If so, explain why. If not, find a minimal sufficient statistic.
 15. Use completeness for the family $N(\theta, 1)$, $\theta \in \mathbb{R}$ to find an essentially unique solution f of the following integral equation:

$$\int f(x)e^{\theta x} dx = \sqrt{2\pi}e^{\theta^2/2}, \quad \theta \in \mathbb{R}.$$

- *16. Let $X_1, \dots, X_n$ be a random sample from an absolutely continuous distribution with density

$$f_\theta(x) = \begin{cases} 2x/\theta^2, & x \in (0, \theta); \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find a one-dimensional sufficient statistic T .
 b) Determine the density of T .
 c) Show directly that T is complete.
 *17. Let $X, X_1, X_2, \dots$ be i.i.d. from an exponential distribution with failure rate λ (introduced in Problem 1.30).
 a) Find the density of $Y = \lambda X$.
 b) Let $\bar{X} = (X_1 + \dots + X_n)/n$. Show that $\bar{X}$ and $(X_1^2 + \dots + X_n^2)/\bar{X}^2$ are independent.
 18. Let $X_1, \dots, X_n$ be independent, with $X_i \sim N(t_i\theta, 1)$, where $t_1, \dots, t_n$ are a sequence of known constants (not all zero).
 a) Show that the least squares estimator $\hat{\theta} = \sum_{i=1}^n t_i X_i / \sum_{i=1}^n t_i^2$ is complete sufficient for the family of joint distributions.
 b) Use Basu's theorem to show that $\hat{\theta}$ and $\sum_{i=1}^n (X_i - t_i \hat{\theta})^2$ are independent.
 19. Let X and Y be independent Poisson variables, X with mean θ , and Y with mean θ^2 , $\theta \in (0, \infty)$.
 a) Find a minimal sufficient statistic for the family of joint distributions.

- b) Is your minimal sufficient statistic complete? Explain.
20. Let $Z_1, \dots, Z_n$ be i.i.d. standard normal variates, and let Z be the random vector formed from these variables. Use Basu's theorem to show that $\|Z\|$ and $Z_1/\|Z\|$ are independent.
21. Let $X_1, \dots, X_n$ be i.i.d. from the uniform distribution on $(0, 1)$, and let $M = \max\{X_1, \dots, X_n\}$. Show that X_1/M and M are independent.
22. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. and absolutely continuous with common density

$$f_{\theta}(x, y) = \begin{cases} 2/\theta^2, & x > 0, y > 0, x + y < \theta; \\ 0, & \text{otherwise.} \end{cases}$$

(This is the density for a uniform distribution on the region inside a triangle in $\mathbb{R}^2$.)

- a) Find a minimal sufficient statistic for the family of joint distributions.
 b) Find the density for your minimal sufficient statistic.
 c) Is the minimal sufficient statistic complete?
23. Suppose X has a geometric distribution with success probability $\theta \in (0, 1)$, Y has a geometric distribution with success probability $2\theta - \theta^2$, and X and Y are independent. Find a minimal sufficient statistic T for the family of joint distributions. Is T complete?
24. Let data X and Y be independent variables with

$$X \sim \text{Binomial}(n, \theta) \text{ and } Y \sim \text{Binomial}(n, \theta^2),$$

with $\theta \in (0, 1)$ an unknown parameter.

- a) Find a minimal sufficient statistic.
 b) Is the minimal sufficient statistic complete? If it is, explain why; if it is not, find a nontrivial function g such that $E_{\theta}g(T) = 0$ for all θ .
25. Let $X_1, \dots, X_n$ be i.i.d. absolutely continuous random variables with common density

$$f_{\theta}(x) = \begin{cases} \theta e^{-\theta x}, & x > 0; \\ 0, & x \leq 0, \end{cases}$$

where $\theta > 0$ is an unknown parameter.

- a) Find the density of θX_i .
 b) Let $X_{(1)} \leq \dots \leq X_{(n)}$ be the order statistics and $\bar{X} = (X_1 + \dots + X_n)/n$ the sample average. Show that $\bar{X}$ and $X_{(1)}/X_{(n)}$ are independent.
26. Two teams play a series of games, stopping as soon as one of the teams has three wins. Assume the games are independent and that the chance the first team wins is an unknown parameter $\theta \in (0, 1)$. Let X denote the number of games the first team wins, and Y the number of games the other team wins.
- a) Find the joint mass function of X and Y .

- b) If our data are X and Y , find a minimal sufficient statistic.
 c) Is the minimal sufficient statistic in part (b) complete? Explain your reasoning.
27. Let $X_1, \dots, X_n$ be i.i.d. from a uniform distribution on $(-\theta, \theta)$, where $\theta > 0$ is an unknown parameter.
- a) Find a minimal sufficient statistic T .
 b) Define

$$V = \frac{\overline{X}}{\max_i X_i - \min_i X_i},$$

where $\overline{X} = (X_1 + \dots + X_n)/n$, the sample average. Show that T and V are independent.

28. Show that if f is defined and bounded on $(-\infty, \infty)$, then f cannot be convex (unless it is constant).
- *29. Find a function on $(0, \infty)$ that is bounded and strictly convex.
- *30. Use convexity to show that the canonical parameter space Ξ of a one-parameter exponential family must be an interval. Specifically, show that if $\eta_0 < \eta < \eta_1$, and if η_0 and η_1 both lie in Ξ , then η must lie in Ξ .
- *31. Let f and g be positive probability densities on $\mathbb{R}$. Use Jensen's inequality to show that

$$\int \log \left(\frac{f(x)}{g(x)} \right) f(x) dx > 0,$$

unless $f = g$ a.e. (If $f = g$, the integral equals zero.) This integral is called the *Kullback–Leibler information*.

32. The geometric mean of a list of positive constants $x_1, \dots, x_n$ is

$$\tilde{x} = (x_1 \times \dots \times x_n)^{1/n},$$

and the arithmetic mean is the average $\bar{x} = (x_1 + \dots + x_n)/n$. Show that $\tilde{x} \leq \bar{x}$.

